

進階程式設計

Advanced Program Design

L2



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上課目標

1. 每個人的位置是固定，請確保你的電腦是可以使用。

Everyone's seat is fixed. Please make sure your computer is functional

2. 本課程授課方式使用程式競賽方式進行

The course is conducted in a format similar to competitive programming.

3. Contest Management System使用Linux系統，換行符號只有LF，而教室電腦使用Windows系統，換行為CR+LF

The Contest Management System uses Linux, where the newline character is LF, while the classroom computers use Windows, which uses CR+LF for newlines.

Method	Percentage(110%)
平時練習 Practice sessions	30%
小考 Quizzes	20%
期中考 Mid-term exam	30%
期末考 Final exam	30%

程式基礎觀念：1.程式寫作習慣-縮排&善用括號

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    int main()
    {
        int i;

        printf("please enter a number:");
        scanf("%d",&i);
        if((i>>2)){
            printf("是2的倍數\n");
        }else{
            printf("是3的倍數\n");
        }
        system("PAUSE");
        return 0;
    }
}
```

```
int main()
{
    char ch;
    printf("請輸入英文");
    scanf("%c", &ch);

    if (ch == 'A'){
        printf("1");
    } if (ch == 'B'){
        printf("1");
    } if (ch == 'C'){
        printf("1");
    } if (ch == 'D'){
        printf("1");
    } if (ch == 'E'){
        printf("1");
    } if (ch == 'F'){
        printf("1");
    } if (ch == 'G'){
        printf("1");
    } if (ch == 'H'){
        printf("1");
    }
}
```

程式基礎觀念：1.程式寫作習慣-縮排

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    int a ;
    printf("任一個數字");
    scanf("%d",&a);
    if(a%2==0){
        printf("2的倍數");
    }
    if(a%3==0){
        printf("3的倍數");
    }
    if(a%2 !=0 && a%2 !=1)
        printf("錯誤\n");

    system("PAUSE");
    return 0;
}
```

沒縮排造成他人解讀困擾



```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]){
    int a;

    printf("任一個數字");
    scanf("%d",&a);
    if(a%2==0){
        printf("2的倍數");
    }
    if(a%3==0){
        printf("3的倍數");
    }
    if(a%2 != 0 && a%2 != 1){
        printf("錯誤\n");
    }

    system("PAUSE");
    return 0;
}
```

程式基礎觀念：1.程式寫作習慣-小括號的重要性

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    int a ;
    printf("任一個數字");
    scanf("%d",&a);
    if(a%2==0){
        printf("2的倍數");
    }
    if(a%3==0){
        printf("3的倍數");
    }
    if(a%2 !=0 && a%2 !=1)
        printf("錯誤\n");

    system("PAUSE");
    return 0;
}
```

善用小括號可
以避免優先權



```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]){
    int a;

    printf("任一個數字");
    scanf("%d",&a);
    if((a%2)==0){
        printf("2的倍數");
    }
    if((a%3)==0){
        printf("3的倍數");
    }
    if(((a%2) != 0) && ((a%2) != 1)){
        printf("錯誤\n");
    }

    system("PAUSE");
    return 0;
}
```


除非熟捻運算子優先表，善用小括號可避免潛在錯誤

C 語言運算子優先權重表

	運算子 Operator	說明 Description	結合順序 Associativity
1	() [] -> . ++ --	Function call Array subscripting Element selection (of struct or union) through pointer Element selection (of struct or union) by reference increment/decrement (suffix) (註1)	左至右(註2)
2	! ~ ++ -- + - * & (type) sizeof	logic NOT bitwise NOT increment/decrement (prefix) unary plus and minus Indirection (dereference, right value) Address-of (left value) type cast Size-of	右至左(註2)
3	* / %	Arithmetic multiplication, division, and remainder	左至右
4	+ -	Arithmetic addition, subtraction	左至右
5	<< >>	Bitwise left shift, right shift	左至右
6	< <= > >=	Relational less than, not greater than Relational greater than, not less than	左至右
7	== !=	Relational equal, not equal	左至右
8	&	Bitwise AND	左至右
9	^	Bitwise XOR	左至右
10		Bitwise OR	左至右
11	&&	Logical AND	左至右
12		Logical OR	左至右
13	?:	Ternary conditional	右至左
14	= += -= *= /= %= &= ^= = <<= >>=	Direct assignment Assignment by sum, difference Assignment by product, quotient, remainder Assignment by bitwise AND, XOR, OR Assignment by bitwise left shift, right shift	右至左
15	,	Comma	左至右

```
int b = 3;
```

```
b = b - 1 + 5;
```

- '='的優先權是14，'-'和'+'的優先權都是4
- '-'和'+'的結合順序是左至右，所以3 - 1之後再去+5
- 最後7賦予給變數b

讀取含有空白鍵的字串問題

Q1: 試寫程式去讀取某字串(除了大小寫英文字母之外，可能含空白鍵、數字、符號等)，並且將其大寫字母轉換成小寫字母，或將小寫字母轉成大寫字母

```
abc123 Hello! C/C++  
ABC123 hELLO! c/c++  
-----  
Process exited after 14.66 seconds with return value 0  
Press any key to continue . . .
```


scanf遇到字串中有空白鍵(Space)的問題

Q1: 試寫程式去讀取某字串(除了大小寫英文字母之外，可能含空白鍵、數字、符號等)，並且將其大寫字母轉換成小寫字母，或將小寫字母轉成大寫字母

➤ 使用gets函數：gets(字元陣列名稱);

```
01 #include <stdio.h>
02 #include <stdlib.h>
03 #include <string.h>
04 int main(int argc, char *argv[]) {
05     int loop;
06     char t, myString[100];
07
08     gets(myString);
09     for(loop=0; loop<strlen(myString); loop++) {
10         t = myString[loop];
11         if((t>='a') && (t<='z')) {
12             printf("%c", t -= ('a'-'A'));
13         } else if((t>='A') && (t<='Z')) {
14             printf("%c", t += ('a'-'A'));
15         } else {
16             printf("%c", t);
17         }
18     }
19     return 0;
20 }
```

➤ scanf("%s",arr)會：

1. 讀取非空白字元直到下一個空白字元，未讀出的字串內容將會遺留在緩衝區
2. 非空白字元之前的空白字元都會被忽略

❑ 空白字元包含Space(0x20)、TAB(0x09)、LF(0x0A)、VT(0x0B)、FF(0x0C)、CR(0x0D)

' '	(0x20)	space (SPC)
'\t'	(0x09)	horizontal tab (TAB)
'\n'	(0x0a)	newline (LF)
'\v'	(0x0b)	vertical tab (VT)
'\f'	(0x0c)	feed (FF)
'\r'	(0x0d)	carriage return (CR)

scanf讀取字元遇到\n的問題

Q2: 撰寫一程式分別讀取數字a和字元b，然後輸出數字a和將字元b轉成16進制的ASCII碼。

```
123
7
=====
123
0x37
-----
Process exited after 1.843 seconds with return value 0
Press any key to continue . . .
```

scanf讀取字元遇到\n的問題

Q2: 撰寫一程式分別讀取數字a和字元b，然後輸出數字a和將字元b轉成16進制的ASCII碼。

Hint: 由於Windows的Enter鍵為CR(0xD)+LF(0xA)，而Linux的Enter鍵為LF(0xA)，因此第2個scanf會讀取到上一個Enter鍵遺留的LF。有下列幾種方法可避免：

```
01 #include <stdio.h>
02 #include <stdlib.h>
03 int main(int argc, char *argv[]) {
04     int num;
05     char t;
06
07     scanf("%d", &num);
08     scanf(" %c", &t);
09     printf("=====\n");
10     printf("%d\n", num);
11     printf("0x%X\n", t);
12     return 0;
13 }
```

前面加一個空白

```
01 #include <stdio.h>
02 #include <stdlib.h>
03 int main(int argc, char *argv[]) {
04     int num;
05     char null, t;
06
07     scanf("%d", &num);
08     scanf("%c%c", &null, &t);
09     printf("=====\n");
10     printf("%d\n", num);
11     printf("0x%X\n", t);
12     return 0;
13 }
```

使用變數null去存取\n

```
01 #include <stdio.h>
02 #include <stdlib.h>
03 int main(int argc, char *argv[]) {
04     int num;
05     char t;
06
07     scanf("%d", &num);
08     scanf("%*[\xd\xa]%c", &t);
09     printf("=====\n");
10     printf("%d\n", num);
11     printf("0x%X\n", t);
12     return 0;
13 }
```

忽略0xA和0xD

The End

